Manual

Real-time Process Data Monitoring

PDV-RPM 8.3

Version 1.0.23049

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1 Overview: Real-Time Process Data Monitoring

Overview

Purpose

The product "PDV-RPM 8.3" includes functions for the online visualization of data on the MOC client. You can visualize the process data that is published via MQTT broker. The logical channels and the collection requests that are actively performed at the machines specify the data that can be displayed.

To visualize data, predefined display instruments and a trend graph are available. You can integrate special layouts into the system. Via the relevant editors, you use these layouts to visualize machines or equipment.

Integration

For the product PDV-RPM 8.3, the following requirements must be fulfilled:
- machine interface for data collection in the Process Communication Controller (PCC),
- data must be published via the MQTT broker,
- the basic PDV-PDM package including collection rules must be available.

Features

This product provides the following functions:

- Permanent display of process data of one or more machines and systems. You can define your own layouts or have them generated via customization.
- Display of real time measured values of a machine using predefined gauge charts. You can select the displayed characteristics in an interactive manner.
- Display of real time measured values of a machine as trend line. The last 50 measured values of a process characteristic are visualized online as continuous trend graph.

2 Configuration PDV Visualization

Purpose

This document describes how to configure layouts for the PDV online visualization on the terminal (AIP 8.2) and the MES Operation Center (MOC).

Requirements

The online visualization via MQTT requires an active license PDV-RPM, AIP-PDV. For the terminal or the MOC, the following requirements must be fulfilled:

MOC:

- Configure the storage path in the HYDRA paths with a key "pdvlay" (Go to: System administration → System settings → Paths).

Terminal:

- The terminal version 8.2.1.1 is required.

Procedure

General files

MPDV provides 5 default layouts as of Hydra service pack 12. The layouts include the following configurations:

- Mixed layout with all available controls
- Layout with line charts (*Charts*)
- Layout with digital displays (*Digital*)
- Layout with level indicator (*Linear*)
- Layout with gauge chart (*Gauge*)

The supply also includes a configuration file "MachineLayoutConfiguration.xml". This file lists the available layouts for the different machines.

Configuration of the layout assignment ("MachineLayoutConfiguration.xml")

Sample configuration:

```xml
<Allocations>
	<LayoutsForAllMachines>
		<Layout id="Global" filename="DefaultLayout.xml">Mix Standard Layout</Layout>
		<Layout id="ALL_1" filename="default_4x4_gauge.xml">Gauge 4x4</Layout>
		<Layout id="ALL_2" filename="default_4x4_digital.xml">Digital 4x4</Layout>
		<Layout id="ALL_3" filename="default_4x4_bar.xml">Bar 4x4</Layout>
		<Layout id="ALL_4" filename="default_4x4_chart.xml">Chart 4x4</Layout>
	</LayoutsForAllMachines>
	<MachineSpecificLayouts>
		<MachineSpecificLayout mnr="xyz">
			<Layouts>
			</Layouts>
		</MachineSpecificLayout>
	</MachineSpecificLayouts>
</Allocations>
```

XML element	Description
LayoutsForAllMachines	Provides a list of layouts that are available with all machines.
MachineSpecificLayouts	Includes the configurations of the layouts for the machines.
MachineSpecificLayout	Provides a list of layouts that are only available with the machines defined in "mnr".
Layout	Describes an individual layout: id: Internal ID for the layout assignment. Must be unambiguous. Filename: File name of the layout. Value: Display name of the layout. e.g. "Mix Standard Layout"

Customization:

- **In the MOC**:
  You can store an individual layout assignment in the HYDRA server when using the MOC. It must be named "MachineLayoutConfiguration_custom.xml".

- **In the terminal**:
  In the terminal, you can create specific layouts for a terminal or a terminal group. The storage of layout configurations is performed according to the default storage of configurations in the terminal. You can find the layouts in the terminal directory in **.\etc\PDV\Layouts**.

The file contents are NOT merged.

Configuration of a layout

Sample configuration:

```
<LayoutGroup>
	<GroupName></GroupName>
	<ItemOrientation>Horizontal</ItemOrientation>
	<Items>
		<LayoutItem>
			<Type>Gauge</Type>
			<MaxValueCount>5</MaxValueCount>
			<Topic>mv/pp/%WORKPLACE%/%PPARAM%</Topic>
			<GuiRefreshTime>2000</GuiRefreshTime>
			<SmoothValueChange>true</SmoothValueChange>
			<MinorTickCount>10</MinorTickCount>
			<MajorTickCount>5</MajorTickCount>
			<Layout>CleanWhite</Layout>
		</LayoutItem>
</LayoutGroup>
```

General, valid for all types of display.

XML element	Description
Type	Specifies the type of display Linear: display of level indicator Digital: digital display Gauge: display of gauge chart Chart: display of trend line If you do not select a display type or the selected type is not found, a gauge chart is automatically

	displayed.
Topic	Specifies the MQTT topic. This specification is mandatory and is currently always "mv/pp/%WORKPLACE%/%PPARAM%". The placeholders %WORKPLACE% and %PPARAM% are replaced with the respective PDV configuration during run time.
GuiRefreshTime	This property modifies the update interval [ms] of the control. Depending on the number of controls (i.e. the process parameters) and the frequence of incoming values, you can increase or decrease the update rate. **Example:** Many values + many process parameters = higher update interval Many values + few process parameters = medium update interval Few values + many process parameters = medium update interval Few values + few process parameters = low update interval **Important:** - The intervals also depend on the hardware equipment of the terminal or computer. - The controls store only the last value that was received. Exception: the chart display (trend line). If further values are received between the updates, the values are discarded and only the most recent value is adopted. The chart saves all incoming values.

Layout	Type of layout. See available layouts.
GroupName	Name of the layout group This name is also displayed in the view.
ItemOrientation	Horizontal: The controls are displayed in a row. Vertical: The controls are displayed in a column.

For charts:

XML element	Description
MaxValueCount	Identifies the maximum number of values to be displayed within the chart.

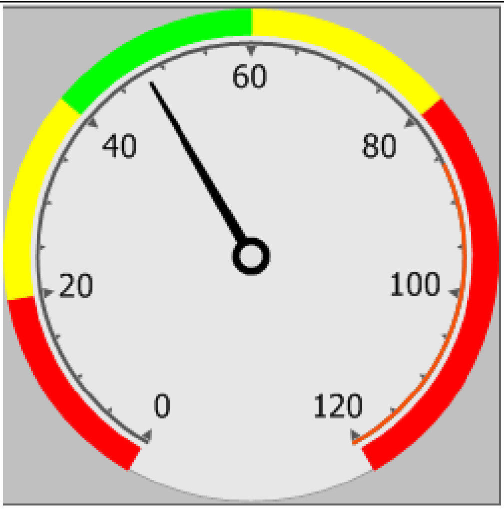
For all display types except chart:

XML element	Description
SmoothValueChange	Enables a smooth transition of the pointer from one value to the next. **Important:** In case of high update rates, you should not use this feature as the smooth transition takes time.
MajorTickCount (not available with *Digital*)*	It is possible to overwrite the MajorTickCounts calculated during run time with a fixed value.
MinorTickCount (not available with *Digital*)*	This property enables a configuration of the number of ticks.

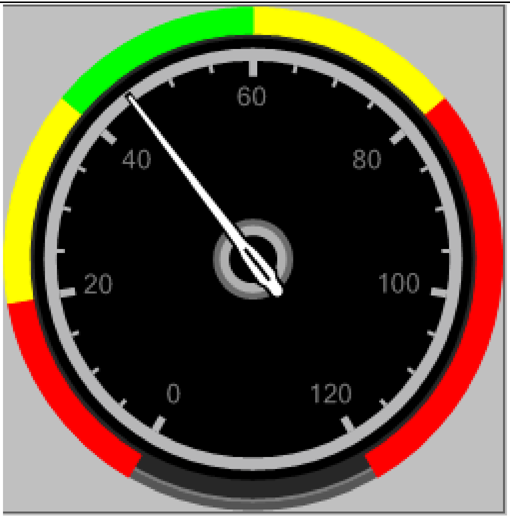
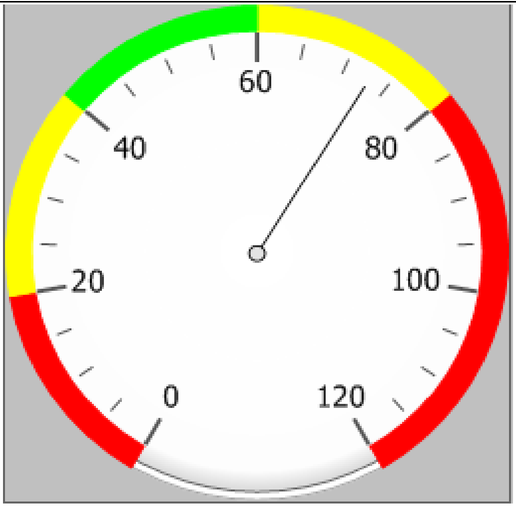
* A tick corresponds to a line in the scale. Therefore, a MajorTick corresponds to a line with a value. The MinorTicks are displayed between the MajorTicks.

Available layouts:

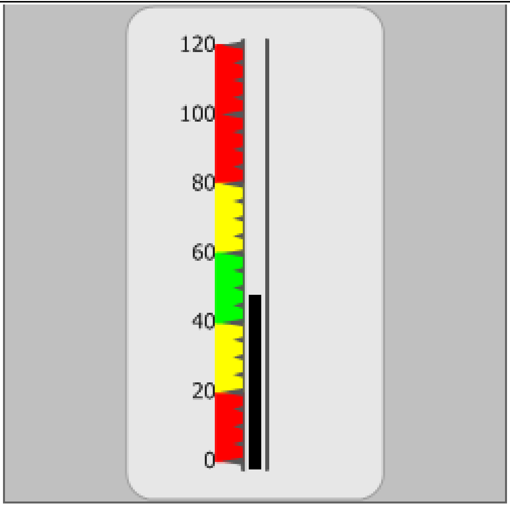
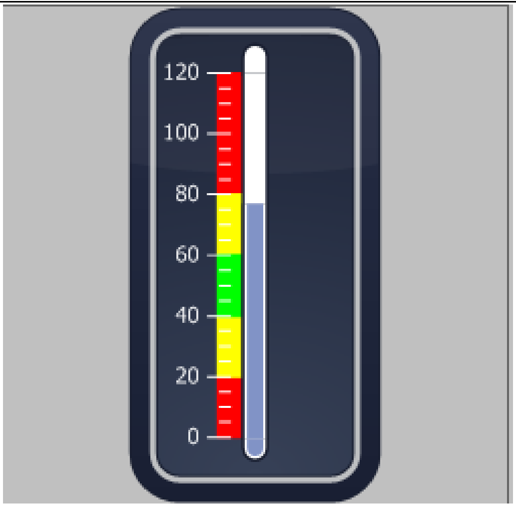
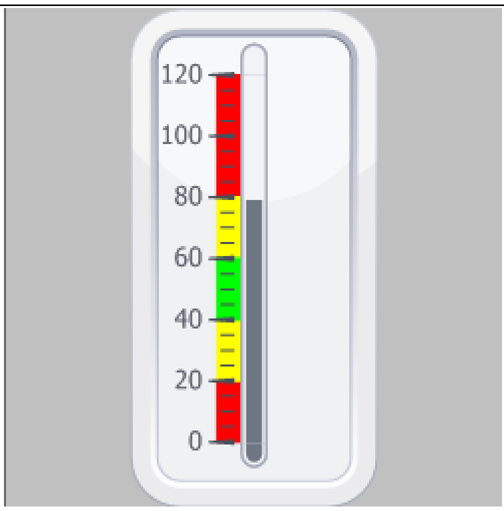
Layout	Display on AIP

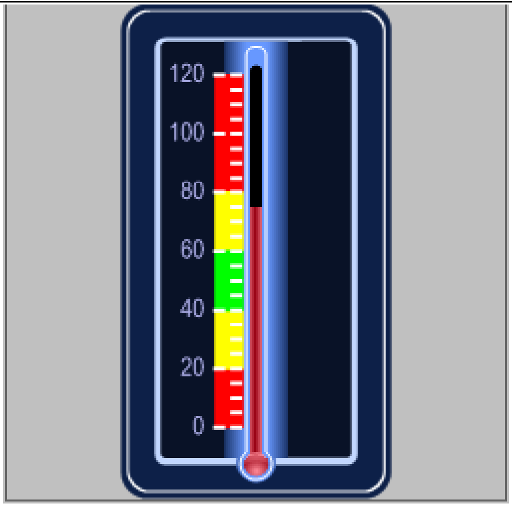
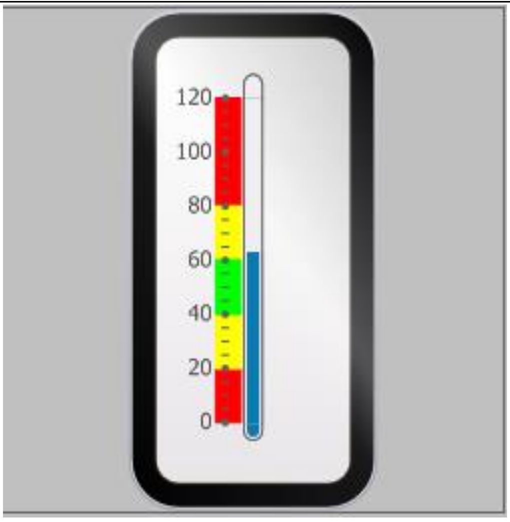
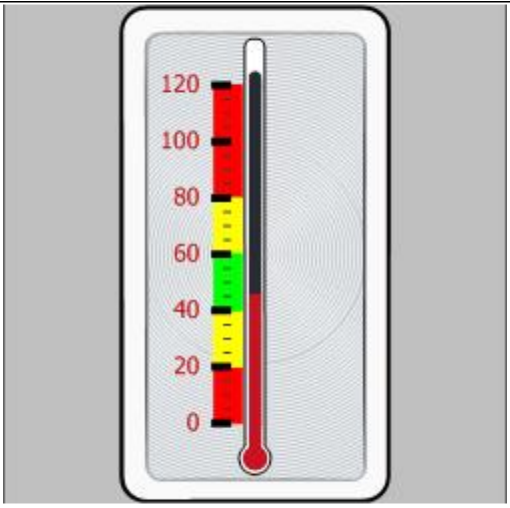
CleanWhite	
Classic	
Clever	

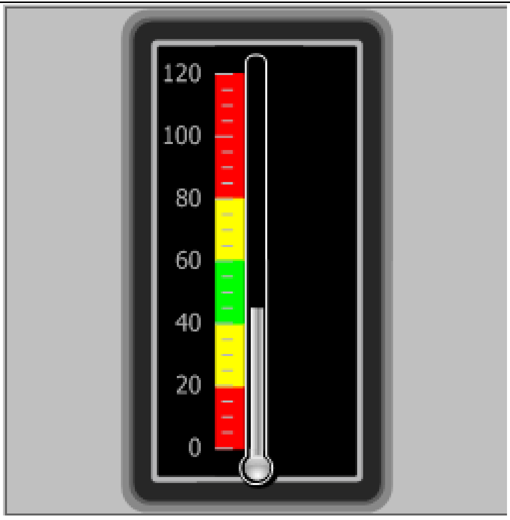
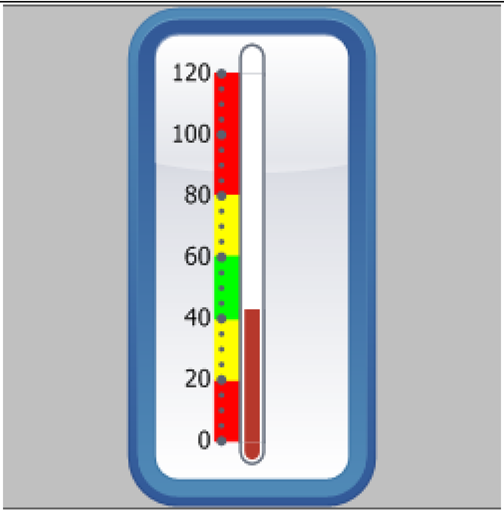
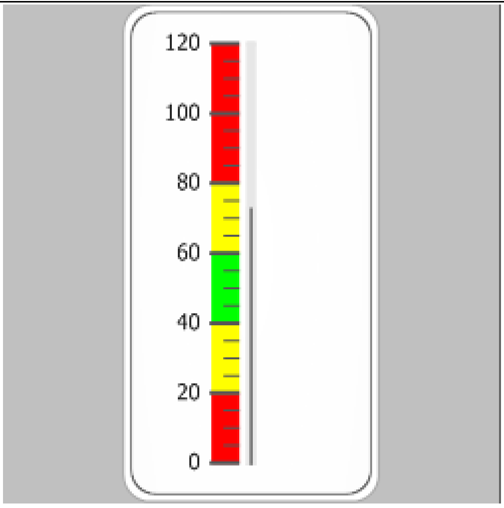
DarkNight	
iStyle	
Retro	

ShiningDark	
Smart	
White	

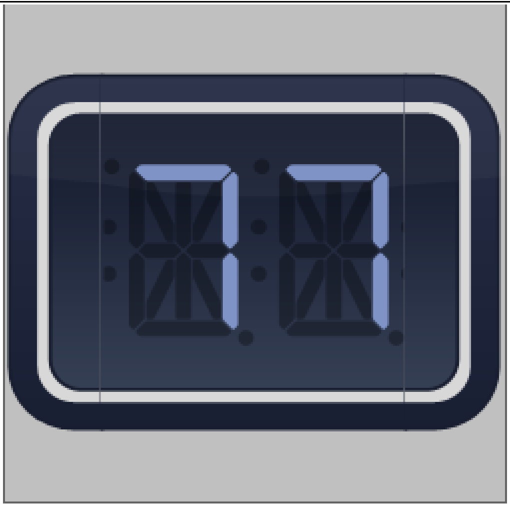
Layout	Display on AIP and MOC

CleanWhite	
Classic	
Clever	

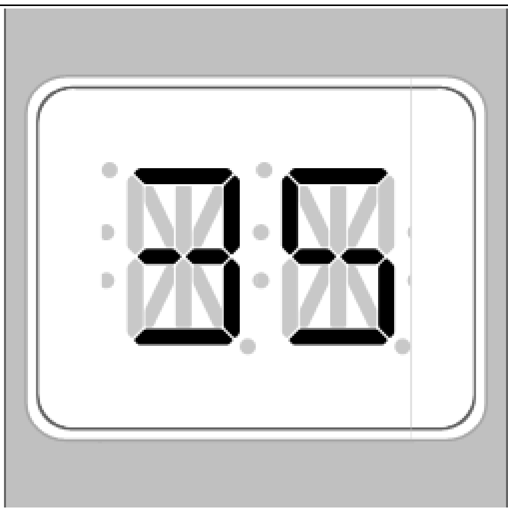
DarkNight	
iStyle	
Retro	

ShiningDark	
Smart	
White	

Layout	Display on AIP and MOC

CleanWhite	
Classic	
Clever	

DarkNight	
iStyle	
Retro	

ShiningDark	
Smart	
White	

3 PDV Online Visualization: MQTT

Overview

The PDV online visualization is based on data provided by the visualization client using the MQTT. MQTT is a client server protocol. Clients send messages to the server ("MQTT broker") with a topic, which classifies the message hierarchically. The clients can subscribe to these topics and the MQTT broker then forwards messages with the topic to the corresponding clients (subscribers).

A message always consists of a topic and the message content (payload).

The PDV online visualization clients available in the HYDRA standard automatically subscribe to the relevant topics and display the process values provided by the MQTT broker.

Implementation notes

- You use the function package if the following applies to you:
  You want to receive and display process data in other tools.

Integration and requirements

This function can only be used if the MQTT broker is set up.

The PCC (PDV Blade) of the collection client provides the data for the online visualization. All data of those process parameters are transmitted to the MQTT broker for online visualization. This setting is made for each characteristic in the PDV data collection rule.

Process data collected with the aid of a web service (REST interface) are not automatically provided to the online visualization via MQTT.

If there is a requirement to provide this data also via MQTT, the collection client must send the data to the MQTT broker at the same time as to the web service.

Topic

In this context a topic stands for a process parameter which is transmitted to the MQTT broker by the collection component.

Topic characters

- Only ASCII characters (avoid non-printable characters)
- Reserved, not allowed characters:
  - Space bar

- Wildcards: + #
- Hierarchy separators: /
- System information: $SYS

Topic:

mv/pp/<workplace>/<process parameter>

Topic	***Value***	***Description/processing***
Mv	mv	Fixed value Abbreviation for measured value
Pp	pp	Fixed value Abbreviation for process parameter
<workplace>	MNR-100	Machine number (URL – encoded)
<process parameter>	Temperature	Process parameter (URL encoded) Note: max. 50 characters

Example MPDV:

Process parameter „Test10“that was collected on machine MDE-100:

Topic	***URL – encoded***
mv/pp/MDE-100/temperature	mv/pp/MDE-100/temperature

/Payload

The payload (message content) is in form of a JSON document.

Example payload:


```json
{
  "type": "mv",
  "ts": "2016-08-16T02:00:00.000Z",
  "id": 101,
  "source": "PCC-10",
  "data":
  {
    "wpl": "MDE-100",
    "id": 1,
    "unit": "°C",
    "name": "Temperature 1",
    "abbr": "Temp",
    "tvalue": 40.0,
    "value": 25.5,
    "max": 120.0,
    "min": 5.0,
    "ul1": 90.0,
    "ul2": 70.0,
    "ll2": 30.5,
    "ll1": 13.0,
    "ts": "2016-08-16T02:00:00.000Z"
  }
}
```


Structure of the JSON document

Attribute	*Value*	*Type*	*Description*	*Mandatory fields*
`type`	Mv	string	Fixed value „mv“ → measured value	X
`ts`	2016-08-16T02:00:00.000Z	timestamp	Time stamp for messages (UTC)	X
`id`	1	int	Message ID Continuous message ID starting with 1	
`source`	PCC-10	string	Sender of the message	
`data`		object	Message object	X

Data object

The data object "data" contains the data of the collected process parameter to be visualized.

Attribute	*Example*	*Type*	*Description*	*Mandatory fields*
`wpl`	MDE-100	string	Machine number	X
`id`	1	int	Channel ID or visualization position	X
`unit`	°C	string	Unit	X
`name`	Temperature S1	string	Process parameter ID	
`abbr`	temp.	string	Characteristic name	
`tvalue`	40.0	decimal	Target value	
`max`	120.0	decimal	Upper PL	
`min`	5.0	decimal	Lower PL	
`ul1`	90.0	decimal	Upper PAL	
`ul2`	70.0	decimal	Upper TL	
`value`	23.45	decimal	Current measured value	X
`ll1`	13.0	decimal	Lower PAL	
`ll2`	30.0	decimal	Lower TL	
`ts`	2016-08-16T02:00:00.000Z	timestamp	PDV 8.3: time stamp of the measured value in ISO8601 format (UTC) PDV 8.2: time stamp in local time of the collection client	X